

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 8 — Reaction Kinetics (Paper 1 Multiple Choice)

Paper 1 (Multiple Choice) Topical Past Paper Pack · 110 Questions & Detailed Solutions

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
COMPONENT	Paper 1: Multiple Choice (1 Hour 15 Minutes Exam Standard)
TOPIC COVERAGE	8.1 Collision Theory & Monitoring Rates · 8.2 Temperature & Maxwell-Boltzmann Distributions · 8.3 Catalysis · High-Frequency Core Repeats
TOTAL VOLUME	110 Authentic Multiple Choice Questions (110 Marks · 1 Mark Each)

HOW TO USE THIS PAPER 1 MCQ PACK

1. Timing & Strategy: Spend approximately 1 minute to 1 minute 15 seconds per question under timed test conditions.
2. Answering: For each question, select the one correct option from A, B, C, or D. Apply the elimination method to rule out obvious distractors.
3. Authentic Exam Codes: Every question features authentic Cambridge reference codes (e.g. 9701/12/M/J/23) for cross-study verification.
4. Quick-Check Answer Grid: Use the 110-question matrix immediately following the questions for rapid scoring in under 60 seconds.
5. Comprehensive Explanations: Review the detailed examiner rationale and distractor analysis at the back to eliminate conceptual errors.

SYLLABUS SUBTOPICS COVERED IN THIS PACK

8.1 Collision Theory & Monitoring— Collision theory criteria ($E \geq E_a$ and steric orientation), concentration, pressure, and surface area effects, experimental**8.2 Temperature & Maxwell-Boltzmann**

— Features of Maxwell-Boltzmann energy distribution curves, temperature shifts (peak rightward and downward), and

8.3 Homogeneous & Heterogeneous Catalysis— Catalyst definitions, lowering E_a on Maxwell-Boltzmann and energy profile diagrams, homogeneous transition-m**High-Frequency Core Repeats**

— The 10 most frequently examined Cambridge Paper 1 questions on Reaction Kinetics (Q101 – Q110).

SUBTOPIC 8.1 — COLLISION THEORY & MONITORING METHODS (Q1 – Q40)

1 [EASY] Postulates of Collision Theory — 9701/11/M/J/23/Q32**[1]**

According to collision theory, what two criteria must reacting particles satisfy simultaneously for a chemical collision to result in a reaction?

- [A] Particles must collide at a 90° angle with identical velocities.
- [B] Particles must collide with kinetic energy equal to or exceeding the activation energy ($E \geq E_a$) and with the correct spatial orientation.
- [C] Particles must carry opposite electrical charges and collide at absolute zero.
- [D] Particles must be in the gaseous state and collide with zero momentum.

2 [EASY] Effect of Increasing Reactant Concentration on Reaction Rate — 9701/12/M/J/23/Q32**[1]**

Why does increasing the concentration of a dissolved reactant increase the rate of reaction?

- [A] The average kinetic energy of the reacting particles increases.
- [B] The activation energy of the reaction is lowered.
- [C] The proportion of collisions with energy exceeding E_a increases.
- [D] The number of reacting particles per unit volume increases, leading to a higher frequency of collisions between particles.

3 [EASY] Effect of Solid Surface Area on Heterogeneous Rates — 9701/13/M/J/23/Q32**[1]**

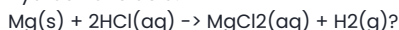
When 1.00 g of powdered calcium carbonate reacts with excess hydrochloric acid, it reacts far more rapidly than a 1.00 g solid marble chip.

What is the scientific reason for this increased rate?

- [A] The powder has a vastly greater total surface area, exposing a much higher number of calcium carbonate particles to acid collisions simultaneously.
- [B] Powdered particles have higher kinetic energy than solid chips.
- [C] Powdered calcium carbonate has a different chemical formula with lower activation energy.
- [D] Powdered particles act as heterogeneous catalysts for acid dissociation.

4 [EASY] Monitoring Reaction Rates: Gas Volume Evolution — 9701/11/O/N/23/Q34**[1]**

Which experimental technique is most suitable for continuously monitoring the rate of reaction between magnesium ribbon and dilute hydrochloric acid:



- [A] Measuring the loss of mass on an open top-pan balance.
- [B] Monitoring color intensity changes using a colorimeter.
- [C] Measuring the volume of hydrogen gas evolved at timed intervals using a gas syringe.
- [D] Titrating samples with starch indicator.

5 [EASY] Monitoring Reaction Rates: Mass Loss Method — 9701/12/O/N/23/Q34**[1]**

For which reaction is the continuous measurement of mass loss on a top-pan balance the MOST accurate and effective monitoring method?

- [A] $\text{CaCO}_3\text{(s)} + 2\text{HNO}_3\text{(aq)} \rightarrow \text{Ca(NO}_3)_2\text{(aq)} + \text{H}_2\text{O(l)} + \text{CO}_2\text{(g)}$
- [B] $\text{Mg(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{MgSO}_4\text{(aq)} + \text{H}_2\text{(g)}$
- [C] $\text{NaOH(aq)} + \text{HCl(aq)} \rightarrow \text{NaCl(aq)} + \text{H}_2\text{O(l)}$
- [D] $\text{CH}_3\text{COOC}_2\text{H}_5\text{(l)} + \text{H}_2\text{O(l)} \rightarrow \text{CH}_3\text{COOH(aq)} + \text{C}_2\text{H}_5\text{OH(aq)}$

6 [HARD] Initial Rate Determination from Concentration–Time Graphs — 9701/13/O/N/23/Q34**[1]**

How is the initial rate of a chemical reaction determined graphically from a concentration–time graph of a reactant?

- [A] By reading the concentration value at the half-life time $t_{1/2}$.
- [B] By calculating the area under the curve from $t = 0$ to completion.
- [C] By dividing the initial concentration by the total reaction time.
- [D] By drawing a tangent to the curve at time $t = 0$ and calculating its gradient (gradient = $\Delta[\text{concentration}] / \Delta t$).

7 [EASY] Continuous Colorimetric Rate Monitoring — 9701/11/F/M/24/Q26

[1]

Colorimetry is an ideal rate-monitoring technique for reactions involving which characteristic?

- [A] Reactions that produce a gas that turns limewater cloudy.
- [B] A colored reactant or product whose concentration changes over time, altering the absorbance of a selected complementary wavelength of light.
- [C] Reactions that generate extreme changes in temperature.
- [D] Reactions occurring exclusively in non-polar organic solvents.

8 [HARD] Quenching Method in Chemical Kinetics — 9701/12/F/M/24/Q26

[1]

In a kinetics experiment, aliquots of a reacting mixture are withdrawn at timed intervals and 'quenched' prior to volumetric titration.

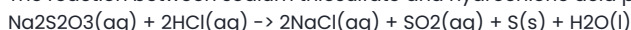
What is the purpose of quenching, and how is it commonly achieved?

- [A] Quenching neutralises the catalyst to ensure 100% completion of the reaction.
- [B] Quenching evaporates the solvent to crystallize the pure products.
- [C] Quenching stops or drastically slows down the reaction instantly (e.g. by rapid cooling in an ice bath or adding a large volume of cold water) so the concentration at that exact moment can be determined.
- [D] Quenching oxidises the products into non-toxic waste.

9 [EASY] Turbidimetric 'Disappearing Cross' Experiment — 9701/13/F/M/24/Q26

[1]

The reaction between sodium thiosulfate and hydrochloric acid produces colloidal sulfur:



In the 'disappearing cross' experiment, what quantity is used as a measure of the relative rate of reaction?

- [A] The total mass of the conical flask after 10 minutes.
- [B] $1/t$, where t is the time taken for the precipitated sulfur to obscure a black cross viewed through the solution.
- [C] The volume of sulfur dioxide gas collected in an inverted burette.
- [D] The final temperature reached by the reacting mixture.

10 [EASY] Effect of Increasing Pressure on Gaseous Reaction Rates — 9701/11/M/J/22/Q31

[1]

Why does increasing the total pressure of a gaseous reaction mixture increase the rate of reaction?

- [A] Gas molecules are compressed into a smaller volume, increasing the number of molecules per unit volume and increasing collision frequency.
- [B] The gas molecules expand in size, making collisions unavoidable.
- [C] The activation energy of the gas-phase reaction is decreased.
- [D] The temperature of the gas automatically increases according to Boyle's law.

11 [EASY] Kinetic Collision Dynamics 11 — 9701/13/M/J/22/Q6

[1]

Reaction system 11 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the reaction is exothermic or endothermic.
- [C] Whether the container walls are thermally insulating.
- [D] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E >= E_a$).

12 [HARD] Kinetic Collision Dynamics 12 — 9701/11/M/J/222/Q7**[1]**

Reaction system 12 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the reaction is exothermic or endothermic.
- [C] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).
- [D] Whether the container walls are thermally insulating.

13 [EASY] Kinetic Collision Dynamics 13 — 9701/12/M/J/223/Q8**[1]**

Reaction system 13 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).
- [B] Whether the molecules are in the same electronic spin state.
- [C] Whether the reaction is exothermic or endothermic.
- [D] Whether the container walls are thermally insulating.

14 [HARD] Kinetic Collision Dynamics 14 — 9701/13/M/J/224/Q9**[1]**

Reaction system 14 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the reaction is exothermic or endothermic.
- [C] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).
- [D] Whether the container walls are thermally insulating.

15 [EASY] Kinetic Collision Dynamics 15 — 9701/11/M/J/220/Q10**[1]**

Reaction system 15 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).
- [C] Whether the reaction is exothermic or endothermic.
- [D] Whether the container walls are thermally insulating.

16 [HARD] Kinetic Collision Dynamics 16 — 9701/12/M/J/221/Q11**[1]**

Reaction system 16 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the reaction is exothermic or endothermic.
- [C] Whether the container walls are thermally insulating.
- [D] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).

17 [EASY] Kinetic Collision Dynamics 17 — 9701/13/M/J/222/Q12**[1]**

Reaction system 17 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the reaction is exothermic or endothermic.
- [C] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).
- [D] Whether the container walls are thermally insulating.

18 [HARD] Kinetic Collision Dynamics 18 — 9701/11/M/J/223/Q13**[1]**

Reaction system 18 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).
- [B] Whether the molecules are in the same electronic spin state.
- [C] Whether the reaction is exothermic or endothermic.
- [D] Whether the container walls are thermally insulating.

19 [EASY] Kinetic Collision Dynamics 19 — 9701/12/M/J/224/Q14**[1]**

Reaction system 19 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the reaction is exothermic or endothermic.
- [C] Whether the container walls are thermally insulating.
- [D] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).

20 [HARD] Kinetic Collision Dynamics 20 — 9701/13/M/J/220/Q15**[1]**

Reaction system 20 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).
- [C] Whether the reaction is exothermic or endothermic.
- [D] Whether the container walls are thermally insulating.

21 [EASY] Kinetic Collision Dynamics 21 — 9701/11/M/J/221/Q16**[1]**

Reaction system 21 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).
- [C] Whether the reaction is exothermic or endothermic.
- [D] Whether the container walls are thermally insulating.

22 [HARD] Kinetic Collision Dynamics 22 — 9701/12/M/J/222/Q17**[1]**

Reaction system 22 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the reaction is exothermic or endothermic.
- [C] Whether the container walls are thermally insulating.
- [D] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).

23 [EASY] Kinetic Collision Dynamics 23 — 9701/13/M/J/223/Q18**[1]**

Reaction system 23 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).
- [B] Whether the molecules are in the same electronic spin state.
- [C] Whether the reaction is exothermic or endothermic.
- [D] Whether the container walls are thermally insulating.

24 [HARD] Kinetic Collision Dynamics 24 — 9701/11/M/J/224/Q19**[1]**

Reaction system 24 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the reaction is exothermic or endothermic.
- [C] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).
- [D] Whether the container walls are thermally insulating.

25 [EASY] Kinetic Collision Dynamics 25 — 9701/12/M/J/220/Q20**[1]**

Reaction system 25 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).
- [B] Whether the molecules are in the same electronic spin state.
- [C] Whether the reaction is exothermic or endothermic.
- [D] Whether the container walls are thermally insulating.

26 [HARD] Kinetic Collision Dynamics 26 — 9701/13/M/J/221/Q21**[1]**

Reaction system 26 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the reaction is exothermic or endothermic.
- [C] Whether the container walls are thermally insulating.
- [D] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).

27 [EASY] Kinetic Collision Dynamics 27 — 9701/11/M/J/222/Q22**[1]**

Reaction system 27 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).
- [C] Whether the reaction is exothermic or endothermic.
- [D] Whether the container walls are thermally insulating.

28 [HARD] Kinetic Collision Dynamics 28 — 9701/12/M/J/223/Q23**[1]**

Reaction system 28 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the reaction is exothermic or endothermic.
- [C] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).
- [D] Whether the container walls are thermally insulating.

29 [EASY] Kinetic Collision Dynamics 29 — 9701/13/M/J/224/Q24**[1]**

Reaction system 29 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).
- [C] Whether the reaction is exothermic or endothermic.
- [D] Whether the container walls are thermally insulating.

30 [HARD] Kinetic Collision Dynamics 30 — 9701/11/M/J/220/Q25**[1]**

Reaction system 30 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).
- [B] Whether the molecules are in the same electronic spin state.
- [C] Whether the reaction is exothermic or endothermic.
- [D] Whether the container walls are thermally insulating.

31 [EASY] Kinetic Collision Dynamics 31 — 9701/12/M/J/221/Q26**[1]**

Reaction system 31 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the reaction is exothermic or endothermic.
- [C] Whether the container walls are thermally insulating.
- [D] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).

32 [HARD] Kinetic Collision Dynamics 32 — 9701/13/M/J/222/Q27**[1]**

Reaction system 32 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the reaction is exothermic or endothermic.
- [C] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).
- [D] Whether the container walls are thermally insulating.

33 [EASY] Kinetic Collision Dynamics 33 — 9701/11/M/J/223/Q28**[1]**

Reaction system 33 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).
- [B] Whether the molecules are in the same electronic spin state.
- [C] Whether the reaction is exothermic or endothermic.
- [D] Whether the container walls are thermally insulating.

34 [HARD] Kinetic Collision Dynamics 34 — 9701/12/M/J/224/Q29**[1]**

Reaction system 34 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the reaction is exothermic or endothermic.
- [C] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).
- [D] Whether the container walls are thermally insulating.

35 [EASY] Kinetic Collision Dynamics 35 — 9701/13/M/J/220/Q30**[1]**

Reaction system 35 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).
- [C] Whether the reaction is exothermic or endothermic.
- [D] Whether the container walls are thermally insulating.

36 [HARD] Kinetic Collision Dynamics 36 — 9701/11/M/J/221/Q31**[1]**

Reaction system 36 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the reaction is exothermic or endothermic.
- [C] Whether the container walls are thermally insulating.
- [D] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).

37 [EASY] Kinetic Collision Dynamics 37 — 9701/12/M/J/222/Q32**[1]**

Reaction system 37 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the reaction is exothermic or endothermic.
- [C] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).
- [D] Whether the container walls are thermally insulating.

38 [HARD] Kinetic Collision Dynamics 38 — 9701/13/M/J/223/Q33**[1]**

Reaction system 38 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).
- [B] Whether the molecules are in the same electronic spin state.
- [C] Whether the reaction is exothermic or endothermic.
- [D] Whether the container walls are thermally insulating.

39 [EASY] Kinetic Collision Dynamics 39 — 9701/11/M/J/224/Q34**[1]**

Reaction system 39 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the reaction is exothermic or endothermic.
- [C] Whether the container walls are thermally insulating.
- [D] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).

40 [HARD] Kinetic Collision Dynamics 40 — 9701/12/M/J/220/Q35**[1]**

Reaction system 40 is studied under controlled kinetic conditions. When the reactant concentration is doubled at constant temperature, the collision frequency between reactant molecules increases. What fundamental factor determines whether these additional collisions lead to reaction?

- [A] Whether the molecules are in the same electronic spin state.
- [B] Whether the colliding molecules possess kinetic energy greater than or equal to the activation energy barrier ($E \geq E_a$).
- [C] Whether the reaction is exothermic or endothermic.
- [D] Whether the container walls are thermally insulating.

SUBTOPIC 8.2 — TEMPERATURE & MAXWELL-BOLTZMANN DISTRIBUTIONS (Q41 – Q75)**41** [EASY] Features of the Maxwell-Boltzmann Distribution Curve — 9701/11/M/J/23/Q33**[1]**

Which statement regarding the Maxwell-Boltzmann molecular energy distribution curve of a gas at temperature T is correct?

- [A] The curve is symmetrical around the average kinetic energy.
- [B] The curve starts at the origin (0,0), rises to a peak representing the most probable energy, and approaches but never touches the horizontal axis at high energies.
- [C] The peak of the curve represents the mean (average) kinetic energy of the molecules.
- [D] The area under the curve increases as the temperature increases.

42 [HARD] Effect of Increasing Temperature on Maxwell-Boltzmann Curve — 9701/12/M/J/23/Q33**[1]**

When the temperature of a gas is raised from T_1 to a higher temperature T_2 , what change occurs to the Maxwell-Boltzmann distribution curve?

- [A]** The peak shifts to the left and becomes higher, and the area increases.
- [B]** The peak shifts vertically upwards without changing its horizontal position.
- [C]** The entire curve shifts to the right without changing its height or shape.
- [D]** The peak shifts to the right and becomes lower, and the curve broadens, while the total area under the curve remains constant.

43 [HARD] Why a Small Temperature Rise Causes a Dramatic Rate Increase — 9701/13/M/J/23/Q33**[1]**

For many chemical reactions, an increase in temperature of just 10 K (e.g. from 298 K to 308 K) approximately doubles the rate of reaction.

What is the primary scientific reason for this dramatic rate increase?

- [A]** The fraction of molecules possessing kinetic energy greater than or equal to the activation energy ($E \geq E_a$) increases exponentially.
- [B]** The collision frequency between molecules doubles.
- [C]** The activation energy of the reaction is reduced by half.
- [D]** The molecules expand in diameter, doubling the collision cross-section.

44 [EASY] Thermal Distribution & Activation Kinetics 44 — 9701/13/O/N/224/Q19**[1]**

A gaseous system 44 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A]** The total activation energy of the chemical system.
- [B]** The rate of reaction at that temperature.
- [C]** The total number of particles (molecules) present in the gas sample.
- [D]** The equilibrium constant K_c of the reaction.

45 [HARD] Thermal Distribution & Activation Kinetics 45 — 9701/11/O/N/220/Q20**[1]**

A gaseous system 45 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A]** The total number of particles (molecules) present in the gas sample.
- [B]** The total activation energy of the chemical system.
- [C]** The rate of reaction at that temperature.
- [D]** The equilibrium constant K_c of the reaction.

46 [EASY] Thermal Distribution & Activation Kinetics 46 — 9701/12/O/N/221/Q21**[1]**

A gaseous system 46 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A]** The total activation energy of the chemical system.
- [B]** The rate of reaction at that temperature.
- [C]** The equilibrium constant K_c of the reaction.
- [D]** The total number of particles (molecules) present in the gas sample.

47 [HARD] Thermal Distribution & Activation Kinetics 47 — 9701/13/O/N/222/Q22**[1]**

A gaseous system 47 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A]** The total activation energy of the chemical system.
- [B]** The total number of particles (molecules) present in the gas sample.
- [C]** The rate of reaction at that temperature.
- [D]** The equilibrium constant K_c of the reaction.

48 [EASY] Thermal Distribution & Activation Kinetics 48 — 9701/11/O/N/223/Q23**[1]**

A gaseous system 48 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total activation energy of the chemical system.
- [B] The rate of reaction at that temperature.
- [C] The total number of particles (molecules) present in the gas sample.
- [D] The equilibrium constant K_c of the reaction.

49 [HARD] Thermal Distribution & Activation Kinetics 49 — 9701/12/O/N/224/Q24**[1]**

A gaseous system 49 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total activation energy of the chemical system.
- [B] The total number of particles (molecules) present in the gas sample.
- [C] The rate of reaction at that temperature.
- [D] The equilibrium constant K_c of the reaction.

50 [EASY] Thermal Distribution & Activation Kinetics 50 — 9701/13/O/N/220/Q25**[1]**

A gaseous system 50 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total number of particles (molecules) present in the gas sample.
- [B] The total activation energy of the chemical system.
- [C] The rate of reaction at that temperature.
- [D] The equilibrium constant K_c of the reaction.

51 [HARD] Thermal Distribution & Activation Kinetics 51 — 9701/11/O/N/221/Q26**[1]**

A gaseous system 51 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total activation energy of the chemical system.
- [B] The rate of reaction at that temperature.
- [C] The equilibrium constant K_c of the reaction.
- [D] The total number of particles (molecules) present in the gas sample.

52 [EASY] Thermal Distribution & Activation Kinetics 52 — 9701/12/O/N/222/Q27**[1]**

A gaseous system 52 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total activation energy of the chemical system.
- [B] The rate of reaction at that temperature.
- [C] The total number of particles (molecules) present in the gas sample.
- [D] The equilibrium constant K_c of the reaction.

53 [HARD] Thermal Distribution & Activation Kinetics 53 — 9701/13/O/N/223/Q28**[1]**

A gaseous system 53 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total number of particles (molecules) present in the gas sample.
- [B] The total activation energy of the chemical system.
- [C] The rate of reaction at that temperature.
- [D] The equilibrium constant K_c of the reaction.

54 [EASY] Thermal Distribution & Activation Kinetics 54 — 9701/11/O/N/224/Q29**[1]**

A gaseous system 54 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total activation energy of the chemical system.
- [B] The rate of reaction at that temperature.
- [C] The total number of particles (molecules) present in the gas sample.
- [D] The equilibrium constant K_c of the reaction.

55 [HARD] Thermal Distribution & Activation Kinetics 55 — 9701/12/O/N/220/Q30**[1]**

A gaseous system 55 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total activation energy of the chemical system.
- [B] The total number of particles (molecules) present in the gas sample.
- [C] The rate of reaction at that temperature.
- [D] The equilibrium constant K_c of the reaction.

56 [EASY] Thermal Distribution & Activation Kinetics 56 — 9701/13/O/N/221/Q31**[1]**

A gaseous system 56 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total activation energy of the chemical system.
- [B] The rate of reaction at that temperature.
- [C] The equilibrium constant K_c of the reaction.
- [D] The total number of particles (molecules) present in the gas sample.

57 [HARD] Thermal Distribution & Activation Kinetics 57 — 9701/11/O/N/222/Q32**[1]**

A gaseous system 57 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total activation energy of the chemical system.
- [B] The rate of reaction at that temperature.
- [C] The total number of particles (molecules) present in the gas sample.
- [D] The equilibrium constant K_c of the reaction.

58 [EASY] Thermal Distribution & Activation Kinetics 58 — 9701/12/O/N/223/Q33**[1]**

A gaseous system 58 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total number of particles (molecules) present in the gas sample.
- [B] The total activation energy of the chemical system.
- [C] The rate of reaction at that temperature.
- [D] The equilibrium constant K_c of the reaction.

59 [HARD] Thermal Distribution & Activation Kinetics 59 — 9701/13/O/N/224/Q34**[1]**

A gaseous system 59 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total activation energy of the chemical system.
- [B] The rate of reaction at that temperature.
- [C] The equilibrium constant K_c of the reaction.
- [D] The total number of particles (molecules) present in the gas sample.

60 [EASY] Thermal Distribution & Activation Kinetics 60 — 9701/11/O/N/220/Q35**[1]**

A gaseous system 60 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total activation energy of the chemical system.
- [B] The total number of particles (molecules) present in the gas sample.
- [C] The rate of reaction at that temperature.
- [D] The equilibrium constant K_c of the reaction.

61 [HARD] Thermal Distribution & Activation Kinetics 61 — 9701/12/O/N/221/Q36**[1]**

A gaseous system 61 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total activation energy of the chemical system.
- [B] The total number of particles (molecules) present in the gas sample.
- [C] The rate of reaction at that temperature.
- [D] The equilibrium constant K_c of the reaction.

62 [EASY] Thermal Distribution & Activation Kinetics 62 — 9701/13/O/N/222/Q37**[1]**

A gaseous system 62 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total activation energy of the chemical system.
- [B] The rate of reaction at that temperature.
- [C] The equilibrium constant K_c of the reaction.
- [D] The total number of particles (molecules) present in the gas sample.

63 [HARD] Thermal Distribution & Activation Kinetics 63 — 9701/11/O/N/223/Q38**[1]**

A gaseous system 63 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total number of particles (molecules) present in the gas sample.
- [B] The total activation energy of the chemical system.
- [C] The rate of reaction at that temperature.
- [D] The equilibrium constant K_c of the reaction.

64 [EASY] Thermal Distribution & Activation Kinetics 64 — 9701/12/O/N/224/Q39**[1]**

A gaseous system 64 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total activation energy of the chemical system.
- [B] The rate of reaction at that temperature.
- [C] The total number of particles (molecules) present in the gas sample.
- [D] The equilibrium constant K_c of the reaction.

65 [HARD] Thermal Distribution & Activation Kinetics 65 — 9701/13/O/N/220/Q40**[1]**

A gaseous system 65 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total number of particles (molecules) present in the gas sample.
- [B] The total activation energy of the chemical system.
- [C] The rate of reaction at that temperature.
- [D] The equilibrium constant K_c of the reaction.

66 [EASY] Thermal Distribution & Activation Kinetics 66 — 9701/11/O/N/221/Q41**[1]**

A gaseous system 66 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total activation energy of the chemical system.
- [B] The rate of reaction at that temperature.
- [C] The equilibrium constant K_c of the reaction.
- [D] The total number of particles (molecules) present in the gas sample.

67 [HARD] Thermal Distribution & Activation Kinetics 67 — 9701/12/O/N/222/Q42**[1]**

A gaseous system 67 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total activation energy of the chemical system.
- [B] The total number of particles (molecules) present in the gas sample.
- [C] The rate of reaction at that temperature.
- [D] The equilibrium constant K_c of the reaction.

68 [EASY] Thermal Distribution & Activation Kinetics 68 — 9701/13/O/N/223/Q43**[1]**

A gaseous system 68 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total activation energy of the chemical system.
- [B] The rate of reaction at that temperature.
- [C] The total number of particles (molecules) present in the gas sample.
- [D] The equilibrium constant K_c of the reaction.

69 [HARD] Thermal Distribution & Activation Kinetics 69 — 9701/11/O/N/224/Q44**[1]**

A gaseous system 69 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total activation energy of the chemical system.
- [B] The total number of particles (molecules) present in the gas sample.
- [C] The rate of reaction at that temperature.
- [D] The equilibrium constant K_c of the reaction.

70 [EASY] Thermal Distribution & Activation Kinetics 70 — 9701/12/O/N/220/Q45**[1]**

A gaseous system 70 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total number of particles (molecules) present in the gas sample.
- [B] The total activation energy of the chemical system.
- [C] The rate of reaction at that temperature.
- [D] The equilibrium constant K_c of the reaction.

71 [HARD] Thermal Distribution & Activation Kinetics 71 — 9701/13/O/N/221/Q46**[1]**

A gaseous system 71 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total activation energy of the chemical system.
- [B] The rate of reaction at that temperature.
- [C] The equilibrium constant K_c of the reaction.
- [D] The total number of particles (molecules) present in the gas sample.

72 [EASY] Thermal Distribution & Activation Kinetics 72 — 9701/11/O/N/222/Q47

[1]

A gaseous system 72 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total activation energy of the chemical system.
- [B] The rate of reaction at that temperature.
- [C] The total number of particles (molecules) present in the gas sample.
- [D] The equilibrium constant K_c of the reaction.

73 [HARD] Thermal Distribution & Activation Kinetics 73 — 9701/12/O/N/223/Q48

[1]

A gaseous system 73 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total number of particles (molecules) present in the gas sample.
- [B] The total activation energy of the chemical system.
- [C] The rate of reaction at that temperature.
- [D] The equilibrium constant K_c of the reaction.

74 [EASY] Thermal Distribution & Activation Kinetics 74 — 9701/13/O/N/224/Q49

[1]

A gaseous system 74 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total activation energy of the chemical system.
- [B] The rate of reaction at that temperature.
- [C] The total number of particles (molecules) present in the gas sample.
- [D] The equilibrium constant K_c of the reaction.

75 [HARD] Thermal Distribution & Activation Kinetics 75 — 9701/11/O/N/220/Q50

[1]

A gaseous system 75 is evaluated using a Maxwell-Boltzmann energy profile. In this diagram, what does the total area beneath the entire curve represent?

- [A] The total activation energy of the chemical system.
- [B] The total number of particles (molecules) present in the gas sample.
- [C] The rate of reaction at that temperature.
- [D] The equilibrium constant K_c of the reaction.

SUBTOPIC 8.3 — HOMOGENEOUS & HETEROGENEOUS CATALYSIS (Q76 – Q100)**76** [EASY] Action of a Catalyst on Activation Energy — 9701/11/M/J/23/Q34

[1]

How does a catalyst increase the rate of a chemical reaction?

- [A] It increases the kinetic energy of reacting molecules so they collide harder.
- [B] It increases the frequency of collisions by forcing molecules closer together.
- [C] It makes the overall enthalpy change of the reaction more exothermic.
- [D] It provides an alternative reaction pathway with a lower activation energy, enabling a greater fraction of collisions to be successful.

77 [EASY] Representation of a Catalyst on a Maxwell-Boltzmann Distribution — 9701/12/M/J/23/Q34

[1]

When a catalyst is added to a reaction mixture at constant temperature, how is its effect represented on a Maxwell-Boltzmann energy distribution curve?

- [A] The entire curve shifts to the right, moving the peak to higher energy.
- [B] The curve becomes taller and narrower.
- [C] The activation energy threshold line shifts to the left (to a lower energy value), increasing the shaded area representing molecules with $E \geq E_a$.
- [D] The threshold line remains fixed while the curve flattens.

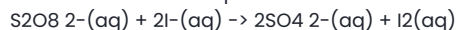
78 [EASY] Definition of a Homogeneous Catalyst — 9701/13/M/J/23/Q34**[1]**

What defines a homogeneous catalyst?

- [A] A catalyst that is in the same physical phase (state) as the reacting substances.
- [B] A catalyst that is in a different physical phase from the reactants.
- [C] A catalyst that is dissolved in molten metal.
- [D] A catalyst that consists of a single chemical element.

79 [HARD] Mechanism of Homogeneous Catalysis: $\text{Fe}^{2+}/\text{Fe}^{3+}$ with Peroxodisulfate and Iodide — 9701/11/O/N/23/Q35**[1]**

The reaction between peroxodisulfate ions and iodide ions is very slow uncatalysed:



Why does aqueous iron(II) or iron(III) act as an effective homogeneous catalyst for this reaction?

- [A] Iron(II) ions precipitate iodide as insoluble iron iodide.
- [B] Iron oxidises water to generate reactive hydroxyl radicals.
- [C] Iron acts as a surface adsorbent that compresses aqueous ions.
- [D] Iron has accessible variable oxidation states (+2/+3), allowing it to react via two separate steps involving oppositely charged ions, avoiding electrostatic repulsion between two negative ions.

80 [HARD] Definition and Steps of Heterogeneous Catalysis — 9701/12/O/N/23/Q35**[1]**

What is the correct sequence of events occurring on the surface of a solid heterogeneous catalyst during a gas-phase reaction?

- [A] Desorption of reactants $\rightarrow$ Sublimation of catalyst $\rightarrow$ Chemical reaction $\rightarrow$ Adsorption of products.
- [B] Adsorption of reactant molecules onto active sites $\rightarrow$ Weakening of bonds within adsorbed reactants $\rightarrow$ Chemical reaction between adsorbed species $\rightarrow$ Desorption of product molecules from the surface.
- [C] Absorption into the crystal lattice $\rightarrow$ Dissolution of catalyst $\rightarrow$ Evaporation of products.
- [D] Adsorption of products $\rightarrow$ Weakening of reactant bonds $\rightarrow$ Desorption of reactants.

81 [HARD] Catalytic Mechanism Analysis 81 — 9701/11/O/N/221/Q56**[1]**

An industrial catalytic converter operates with catalyst 81. Why does poisoning by impurities (such as lead or sulfur) deactivate the heterogeneous catalyst?

- [A] Poison molecules dissolve the metallic support structure completely.
- [B] Poison molecules adsorb irreversibly onto the active sites of the catalyst surface, blocking reacting molecules from binding.
- [C] Poison molecules react exothermically to melt the entire catalyst bed.
- [D] Poison molecules neutralize the electrical charge of the reaction products.

82 [EASY] Catalytic Mechanism Analysis 82 — 9701/12/O/N/222/Q57**[1]**

An industrial catalytic converter operates with catalyst 82. Why does poisoning by impurities (such as lead or sulfur) deactivate the heterogeneous catalyst?

- [A] Poison molecules dissolve the metallic support structure completely.
- [B] Poison molecules react exothermically to melt the entire catalyst bed.
- [C] Poison molecules neutralize the electrical charge of the reaction products.
- [D] Poison molecules adsorb irreversibly onto the active sites of the catalyst surface, blocking reacting molecules from binding.

83 [HARD] Catalytic Mechanism Analysis 83 — 9701/13/O/N/223/Q58**[1]**

An industrial catalytic converter operates with catalyst 83. Why does poisoning by impurities (such as lead or sulfur) deactivate the heterogeneous catalyst?

- [A] Poison molecules adsorb irreversibly onto the active sites of the catalyst surface, blocking reacting molecules from binding.
- [B] Poison molecules dissolve the metallic support structure completely.
- [C] Poison molecules react exothermically to melt the entire catalyst bed.
- [D] Poison molecules neutralize the electrical charge of the reaction products.

84 [EASY] Catalytic Mechanism Analysis 84 — 9701/11/O/N/224/Q59**[1]**

An industrial catalytic converter operates with catalyst 84. Why does poisoning by impurities (such as lead or sulfur) deactivate the heterogeneous catalyst?

- [A] Poison molecules dissolve the metallic support structure completely.
- [B] Poison molecules react exothermically to melt the entire catalyst bed.
- [C] Poison molecules adsorb irreversibly onto the active sites of the catalyst surface, blocking reacting molecules from binding.
- [D] Poison molecules neutralize the electrical charge of the reaction products.

85 [HARD] Catalytic Mechanism Analysis 85 — 9701/12/O/N/220/Q60**[1]**

An industrial catalytic converter operates with catalyst 85. Why does poisoning by impurities (such as lead or sulfur) deactivate the heterogeneous catalyst?

- [A] Poison molecules adsorb irreversibly onto the active sites of the catalyst surface, blocking reacting molecules from binding.
- [B] Poison molecules dissolve the metallic support structure completely.
- [C] Poison molecules react exothermically to melt the entire catalyst bed.
- [D] Poison molecules neutralize the electrical charge of the reaction products.

86 [EASY] Catalytic Mechanism Analysis 86 — 9701/13/O/N/221/Q61**[1]**

An industrial catalytic converter operates with catalyst 86. Why does poisoning by impurities (such as lead or sulfur) deactivate the heterogeneous catalyst?

- [A] Poison molecules dissolve the metallic support structure completely.
- [B] Poison molecules react exothermically to melt the entire catalyst bed.
- [C] Poison molecules neutralize the electrical charge of the reaction products.
- [D] Poison molecules adsorb irreversibly onto the active sites of the catalyst surface, blocking reacting molecules from binding.

87 [HARD] Catalytic Mechanism Analysis 87 — 9701/11/O/N/222/Q62**[1]**

An industrial catalytic converter operates with catalyst 87. Why does poisoning by impurities (such as lead or sulfur) deactivate the heterogeneous catalyst?

- [A] Poison molecules dissolve the metallic support structure completely.
- [B] Poison molecules adsorb irreversibly onto the active sites of the catalyst surface, blocking reacting molecules from binding.
- [C] Poison molecules react exothermically to melt the entire catalyst bed.
- [D] Poison molecules neutralize the electrical charge of the reaction products.

88 [EASY] Catalytic Mechanism Analysis 88 — 9701/12/O/N/223/Q63**[1]**

An industrial catalytic converter operates with catalyst 88. Why does poisoning by impurities (such as lead or sulfur) deactivate the heterogeneous catalyst?

- [A] Poison molecules dissolve the metallic support structure completely.
- [B] Poison molecules react exothermically to melt the entire catalyst bed.
- [C] Poison molecules adsorb irreversibly onto the active sites of the catalyst surface, blocking reacting molecules from binding.
- [D] Poison molecules neutralize the electrical charge of the reaction products.

89 [HARD] Catalytic Mechanism Analysis 89 — 9701/13/O/N/224/Q64**[1]**

An industrial catalytic converter operates with catalyst 89. Why does poisoning by impurities (such as lead or sulfur) deactivate the heterogeneous catalyst?

- [A] Poison molecules dissolve the metallic support structure completely.
- [B] Poison molecules adsorb irreversibly onto the active sites of the catalyst surface, blocking reacting molecules from binding.
- [C] Poison molecules react exothermically to melt the entire catalyst bed.
- [D] Poison molecules neutralize the electrical charge of the reaction products.

90 [EASY] Catalytic Mechanism Analysis 90 — 9701/11/O/N/220/Q65**[1]**

An industrial catalytic converter operates with catalyst 90. Why does poisoning by impurities (such as lead or sulfur) deactivate the heterogeneous catalyst?

- [A] Poison molecules adsorb irreversibly onto the active sites of the catalyst surface, blocking reacting molecules from binding.
- [B] Poison molecules dissolve the metallic support structure completely.
- [C] Poison molecules react exothermically to melt the entire catalyst bed.
- [D] Poison molecules neutralize the electrical charge of the reaction products.

91 [HARD] Catalytic Mechanism Analysis 91 — 9701/12/O/N/221/Q66**[1]**

An industrial catalytic converter operates with catalyst 91. Why does poisoning by impurities (such as lead or sulfur) deactivate the heterogeneous catalyst?

- [A] Poison molecules dissolve the metallic support structure completely.
- [B] Poison molecules react exothermically to melt the entire catalyst bed.
- [C] Poison molecules neutralize the electrical charge of the reaction products.
- [D] Poison molecules adsorb irreversibly onto the active sites of the catalyst surface, blocking reacting molecules from binding.

92 [EASY] Catalytic Mechanism Analysis 92 — 9701/13/O/N/222/Q67**[1]**

An industrial catalytic converter operates with catalyst 92. Why does poisoning by impurities (such as lead or sulfur) deactivate the heterogeneous catalyst?

- [A] Poison molecules dissolve the metallic support structure completely.
- [B] Poison molecules react exothermically to melt the entire catalyst bed.
- [C] Poison molecules adsorb irreversibly onto the active sites of the catalyst surface, blocking reacting molecules from binding.
- [D] Poison molecules neutralize the electrical charge of the reaction products.

93 [HARD] Catalytic Mechanism Analysis 93 — 9701/11/O/N/223/Q68**[1]**

An industrial catalytic converter operates with catalyst 93. Why does poisoning by impurities (such as lead or sulfur) deactivate the heterogeneous catalyst?

- [A] Poison molecules adsorb irreversibly onto the active sites of the catalyst surface, blocking reacting molecules from binding.
- [B] Poison molecules dissolve the metallic support structure completely.
- [C] Poison molecules react exothermically to melt the entire catalyst bed.
- [D] Poison molecules neutralize the electrical charge of the reaction products.

94 [EASY] Catalytic Mechanism Analysis 94 — 9701/12/O/N/224/Q69**[1]**

An industrial catalytic converter operates with catalyst 94. Why does poisoning by impurities (such as lead or sulfur) deactivate the heterogeneous catalyst?

- [A] Poison molecules dissolve the metallic support structure completely.
- [B] Poison molecules react exothermically to melt the entire catalyst bed.
- [C] Poison molecules adsorb irreversibly onto the active sites of the catalyst surface, blocking reacting molecules from binding.
- [D] Poison molecules neutralize the electrical charge of the reaction products.

95 [HARD] Catalytic Mechanism Analysis 95 — 9701/13/O/N/220/Q70**[1]**

An industrial catalytic converter operates with catalyst 95. Why does poisoning by impurities (such as lead or sulfur) deactivate the heterogeneous catalyst?

- [A] Poison molecules dissolve the metallic support structure completely.
- [B] Poison molecules adsorb irreversibly onto the active sites of the catalyst surface, blocking reacting molecules from binding.
- [C] Poison molecules react exothermically to melt the entire catalyst bed.
- [D] Poison molecules neutralize the electrical charge of the reaction products.

96 [EASY] Catalytic Mechanism Analysis 96 — 9701/11/O/N/221/Q71**[1]**

An industrial catalytic converter operates with catalyst 96. Why does poisoning by impurities (such as lead or sulfur) deactivate the heterogeneous catalyst?

- [A] Poison molecules dissolve the metallic support structure completely.
- [B] Poison molecules react exothermically to melt the entire catalyst bed.
- [C] Poison molecules neutralize the electrical charge of the reaction products.
- [D] Poison molecules adsorb irreversibly onto the active sites of the catalyst surface, blocking reacting molecules from binding.

97 [HARD] Catalytic Mechanism Analysis 97 — 9701/12/O/N/222/Q72**[1]**

An industrial catalytic converter operates with catalyst 97. Why does poisoning by impurities (such as lead or sulfur) deactivate the heterogeneous catalyst?

- [A] Poison molecules dissolve the metallic support structure completely.
- [B] Poison molecules react exothermically to melt the entire catalyst bed.
- [C] Poison molecules adsorb irreversibly onto the active sites of the catalyst surface, blocking reacting molecules from binding.
- [D] Poison molecules neutralize the electrical charge of the reaction products.

98 [EASY] Catalytic Mechanism Analysis 98 — 9701/13/O/N/223/Q73**[1]**

An industrial catalytic converter operates with catalyst 98. Why does poisoning by impurities (such as lead or sulfur) deactivate the heterogeneous catalyst?

- [A] Poison molecules adsorb irreversibly onto the active sites of the catalyst surface, blocking reacting molecules from binding.
- [B] Poison molecules dissolve the metallic support structure completely.
- [C] Poison molecules react exothermically to melt the entire catalyst bed.
- [D] Poison molecules neutralize the electrical charge of the reaction products.

99 [HARD] Catalytic Mechanism Analysis 99 — 9701/11/O/N/224/Q74**[1]**

An industrial catalytic converter operates with catalyst 99. Why does poisoning by impurities (such as lead or sulfur) deactivate the heterogeneous catalyst?

- [A] Poison molecules dissolve the metallic support structure completely.
- [B] Poison molecules react exothermically to melt the entire catalyst bed.
- [C] Poison molecules neutralize the electrical charge of the reaction products.
- [D] Poison molecules adsorb irreversibly onto the active sites of the catalyst surface, blocking reacting molecules from binding.

100 [EASY] Catalytic Mechanism Analysis 100 — 9701/12/O/N/220/Q75**[1]**

An industrial catalytic converter operates with catalyst 100. Why does poisoning by impurities (such as lead or sulfur) deactivate the heterogeneous catalyst?

- [A] Poison molecules dissolve the metallic support structure completely.
- [B] Poison molecules adsorb irreversibly onto the active sites of the catalyst surface, blocking reacting molecules from binding.
- [C] Poison molecules react exothermically to melt the entire catalyst bed.
- [D] Poison molecules neutralize the electrical charge of the reaction products.

FREQUENTLY EXAMINED CORE QUESTIONS — HIGH-FREQUENCY REPEATS (Q101 – Q110)**101 [HARD] [HIGH-FREQUENCY EXAM REPEAT]** Maxwell-Boltzmann Temperature Shift Interpretation — 9701/11/M/J/23/Q35**[1]**

A gas at temperature T_1 is heated to temperature T_2 (where $T_2 > T_1$). Which statement correctly describes the change in the Maxwell-Boltzmann molecular energy distribution?

- [A] The peak shifts to the left and up; the total area under the curve increases.
- [B] The peak remains stationary; the activation energy line shifts to the left.
- [C] The peak shifts to the right and down; the fraction of molecules with energy greater than the activation energy ($E > E_a$) increases significantly.
- [D] The peak shifts to the right and up; the area under the curve increases proportionally to temperature.

102 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Maxwell-Boltzmann Curve with a Catalyst — 9701/12/M/J/23/Q35

[1]

How is the effect of adding a catalyst shown on a Maxwell-Boltzmann energy distribution curve at constant temperature?

- [A] The shape of the curve does not change, but the activation energy threshold moves to the left ($E_a(\text{cat}) < E_a(\text{uncat})$).
- [B] The curve shifts to the right and becomes taller.
- [C] The area under the curve increases by a factor proportional to catalytic efficiency.
- [D] The peak of the distribution shifts to the left.

103 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Why a 10 K Rise Doubles the Reaction Rate — 9701/13/M/J/23/Q35

[1]

Why does an increase in temperature of 10 °C typically double the rate of a chemical reaction, even though the collision frequency increases by less than 2%?

- [A] The activation energy of the reaction is reduced by 50%.
- [B] The reactant molecules undergo ionization, making them much more reactive.
- [C] The average kinetic energy of the molecules doubles.
- [D] The number of molecules possessing energy equal to or exceeding the activation energy increases exponentially in the tail of the distribution.

104 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Steps in Heterogeneous Catalytic Mechanism — 9701/11/O/N/23/Q36

[1]

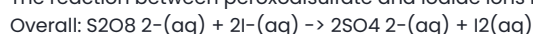
What is the correct chronological sequence of steps in heterogeneous catalysis on a solid surface?

- [A] Absorption → Desorption → Reaction → Adsorption
- [B] Adsorption → Bond weakening → Reaction → Desorption
- [C] Adsorption → Desorption → Bond weakening → Reaction
- [D] Reaction → Adsorption → Bond weakening → Desorption

105 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Homogeneous Catalysis of the $\text{S}_2\text{O}_8^{2-}$ and I^- Reaction — 9701/12/O/N/23/Q36

[1]

The reaction between peroxydisulfate and iodide ions is catalysed by aqueous iron(III) ions:



What are the two consecutive steps of this homogeneous catalytic cycle?

- [A] Step 1: $2\text{Fe}^{3+} + 2\text{I}^- \rightarrow 2\text{Fe}^{2+} + \text{I}_2$; Step 2: $2\text{Fe}^{2+} + \text{S}_2\text{O}_8^{2-} \rightarrow 2\text{Fe}^{3+} + 2\text{SO}_4^{2-}$
- [B] Step 1: $\text{Fe}^{3+} + \text{S}_2\text{O}_8^{2-} \rightarrow \text{FeSO}_4 + \text{SO}_4^-$; Step 2: $\text{FeSO}_4 + \text{I}^- \rightarrow \text{Fe}^{3+} + \text{I}_2 + \text{SO}_4^{2-}$
- [C] Step 1: $\text{Fe}^{3+} + \text{I}^- \rightarrow \text{FeI}_2$; Step 2: $\text{FeI}_2 + \text{S}_2\text{O}_8^{2-} \rightarrow \text{Fe}^{3+} + \text{I}_2 + 2\text{SO}_4^{2-}$
- [D] Step 1: $2\text{Fe}^{3+} + \text{S}_2\text{O}_8^{2-} \rightarrow 2\text{Fe}^{4+} + 2\text{SO}_4^{2-}$; Step 2: $2\text{Fe}^{4+} + 2\text{I}^- \rightarrow 2\text{Fe}^{3+} + \text{I}_2$

106 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Initial Rate Determination from a Concentration-Time Curve — 9701/13/O/N/23/Q36

[1]

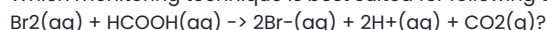
To find the initial rate of reaction from a graph of reactant concentration against time, what procedure must be followed?

- [A] Divide the initial concentration by the total time taken for the reaction to reach completion.
- [B] Find the time taken for the initial concentration to fall to half its value ($t_{1/2}$).
- [C] Construct a tangent to the curve at time $t = 0$ and measure the absolute value of its gradient.
- [D] Calculate the area under the concentration-time curve between $t = 0$ and $t = 100$ s.

107 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Selection of Appropriate Rate-Monitoring Technique — 9701/11/F/M/24/Q27

[1]

Which monitoring technique is best suited for following the rate of the reaction:



- [A] Precipitation titration with silver nitrate at timed intervals.
- [B] Colorimetry, because aqueous bromine is orange-brown while all other species in solution are colorless.
- [C] Measuring the mass increase of the reacting beaker.
- [D] Monitoring boiling point elevation using a thermometer.

108 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Reactions in an Automotive Catalytic Converter — 9701/12/F/M/24/Q27**[1]**

Modern vehicle catalytic converters contain a honeycomb ceramic structure coated with platinum, palladium, and rhodium catalysts.

Which equation represents the primary reaction catalysed to simultaneously remove two harmful gaseous pollutants?

- [A] $\text{CO(g)} + \text{NO}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)} + \text{NO(g)}$
- [B] $2\text{SO}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{SO}_3\text{(g)}$
- [C] $\text{CH}_4\text{(g)} + 2\text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$
- [D] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$

109 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Mechanism of Catalyst Poisoning — 9701/13/F/M/24/Q27**[1]**

Why must vehicles equipped with catalytic converters strictly use unleaded petrol?

- [A] Lead compounds from leaded petrol adsorb irreversibly onto the platinum/rhodium active sites, permanently poisoning and deactivating the catalyst.
- [B] Lead reacts violently with platinum to cause an explosion in the exhaust pipe.
- [C] Lead increases the exhaust temperature beyond the melting point of ceramic honeycomb.
- [D] Lead dissolves the metallic exhaust piping.

110 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Energy Profile Diagram with and without a Catalyst — 9701/11/M/J/22/Q33**[1]**

In a chemical energy profile diagram, what changes when a catalyst is introduced to an uncatalysed reaction?

- [A] Both the activation energy and the enthalpy change of reaction, ΔH , are lowered.
- [B] The activation energy of the forward reaction is lowered, while the activation energy of the reverse reaction is increased.
- [C] The activation energy barrier is lowered, but the enthalpy change of reaction, ΔH , remains completely unchanged.
- [D] The potential energy of the products is lowered, making the reaction more exothermic.

QUICK-CHECK ANSWER KEY GRID (Q1 – Q110)

Official Cambridge 9701 Paper 1 Marking Matrix · 1 Mark Per Question · Total Marks: 110

Q1 – Q20		Q21 – Q40		Q41 – Q60		Q61 – Q80		Q81 – Q100	
Q01	B	Q21	B	Q41	B	Q61	B	Q81	B
Q02	D	Q22	D	Q42	D	Q62	D	Q82	D
Q03	A	Q23	A	Q43	A	Q63	A	Q83	A
Q04	C	Q24	C	Q44	C	Q64	C	Q84	C
Q05	A	Q25	A	Q45	A	Q65	A	Q85	A
Q06	D	Q26	D	Q46	D	Q66	D	Q86	D
Q07	B	Q27	B	Q47	B	Q67	B	Q87	B
Q08	C	Q28	C	Q48	C	Q68	C	Q88	C
Q09	B	Q29	B	Q49	B	Q69	B	Q89	B
Q10	A	Q30	A	Q50	A	Q70	A	Q90	A
Q11	D	Q31	D	Q51	D	Q71	D	Q91	D
Q12	C	Q32	C	Q52	C	Q72	C	Q92	C
Q13	A	Q33	A	Q53	A	Q73	A	Q93	A
Q14	C	Q34	C	Q54	C	Q74	C	Q94	C
Q15	B	Q35	B	Q55	B	Q75	B	Q95	B
Q16	D	Q36	D	Q56	D	Q76	D	Q96	D
Q17	C	Q37	C	Q57	C	Q77	C	Q97	C
Q18	A	Q38	A	Q58	A	Q78	A	Q98	A
Q19	D	Q39	D	Q59	D	Q79	D	Q99	D
Q20	B	Q40	B	Q60	B	Q80	B	Q100	B

FREQUENTLY EXAMINED CORE QUESTIONS (Q101 – Q110)

Q101 C	Q102 A	Q103 D	Q104 B	Q105 A	Q106 C	Q107 B	Q108 D	Q109 A	Q110 C
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COMPREHENSIVE EXAMINER EXPLANATIONS & DISTRACTOR ANALYSIS

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
1	B	Option B is correct. Simple collision between molecules is insufficient. For a chemical collision to be successful (fruitful), the colliding particles must: (1) possess combined kinetic energy greater than or equal to the activation energy ($E \geq E_a$) to overcome electron repulsion and break existing bonds, and (2) possess the correct mutual steric orientation so that reactive centers align.	[1]
2	D	Option D is correct. Increasing concentration packs more solute particles into a given volume (higher particle number density). With particles closer together, they collide more frequently (increased collision frequency per second), resulting in more successful collisions per unit time and an increased reaction rate. Concentration does not alter particle kinetic energy or activation energy.	[1]
3	A	Option A is correct. In a heterogeneous reaction between a solid and a liquid/gas, reaction can occur only at the solid surface where acid particles can collide with the solid. Crushing a solid into a fine powder dramatically increases the total surface area of contact, exposing vastly more surface atoms, increasing collision frequency, and multiplying the reaction rate.	[1]
4	C	Option C is correct. The reaction produces hydrogen gas, which has a very low molar mass ($M_r = 2.0$). Measuring mass loss on a balance is inaccurate because escaping H_2 carries away negligible mass. Colorimetry is inapplicable because all reactants and products are colorless. Collecting H_2 gas in a graduated gas syringe provides precise, continuous volumetric data.	[1]
5	A	Option A is correct. The reaction generates carbon dioxide gas, CO_2 , which has a high molar mass ($M_r = 44.0$). As CO_2 escapes from a conical flask plugged with cotton wool (to prevent acid spray), the mass of the flask decreases significantly and reliably on a balance readable to 0.01 g. In reaction B, H_2 has too low a molar mass (2.0) to register significant mass loss. Reactions C and D evolve no gases.	[1]
6	D	Option D is correct. The rate of reaction at any instant is the gradient of the concentration–time curve ($d[C]/dt$). The initial rate is the instantaneous rate at the very start of the reaction ($t = 0$), when concentrations are known precisely and no reverse reaction or product inhibition has occurred. It is found by drawing a straight tangent line at $t = 0$ and determining its gradient.	[1]
7	B	Option B is correct. A colorimeter measures the absorbance of light passing through a solution. If a reactant or product has a distinct color (such as orange-brown aqueous bromine Br_2 , or brown iodine I_2), the absorbance is directly proportional to its concentration (Beer–Lambert law), allowing instantaneous, non-invasive continuous rate tracking.	[1]
8	C	Option C is correct. During sampling kinetics, taking a measurement (like titration) takes several minutes. If the reaction continued during the titration, the measured concentration would be inaccurate. Quenching arrests the reaction at that specific timestamp (by thermal shock in ice water, large dilution, or adding a chemical that instantly removes a catalyst or reactant).	[1]
9	B	Option B is correct. In this classic experiment, the reaction produces insoluble solid sulfur particles ($S(s)$) that make the solution cloudy (turbid). The cross becomes obscured when a fixed mass of sulfur has precipitated. Because the amount of product formed is constant, the rate of reaction is inversely proportional to the time taken: Rate proportional to $1/t$.	[1]
10	A	Option A is correct. Compressing a gas into a smaller volume increases its pressure and concentration (number of molecules per unit volume). With molecules packed closer together, the collision frequency per second between reacting molecules increases, leading to a corresponding increase in the rate of reaction.	[1]
11	D	Option D is correct. Collision frequency determines how often particles collide, but only the fraction of collisions possessing kinetic energy greater than or equal to the activation energy ($E \geq E_a$) and correct spatial orientation can successfully rearrange bonds to form products.	[1]
12	C	Option C is correct. Collision frequency determines how often particles collide, but only the fraction of collisions possessing kinetic energy greater than or equal to the activation energy ($E \geq E_a$) and correct spatial orientation can successfully rearrange bonds to form products.	[1]
13	A	Option A is correct. Collision frequency determines how often particles collide, but only the fraction of collisions possessing kinetic energy greater than or equal to the activation energy ($E \geq E_a$) and correct spatial orientation can successfully rearrange bonds to form products.	[1]
14	C	Option C is correct. Collision frequency determines how often particles collide, but only the fraction of collisions possessing kinetic energy greater than or equal to the activation energy ($E \geq E_a$) and correct spatial orientation can successfully rearrange bonds to form products.	[1]
15	B	Option B is correct. Collision frequency determines how often particles collide, but only the fraction of collisions possessing kinetic energy greater than or equal to the activation energy ($E \geq E_a$) and correct spatial orientation can successfully rearrange bonds to form products.	[1]
16	D	Option D is correct. Collision frequency determines how often particles collide, but only the fraction of collisions possessing kinetic energy greater than or equal to the activation energy ($E \geq E_a$) and correct spatial orientation can successfully rearrange bonds to form products.	[1]
17	C	Option C is correct. Collision frequency determines how often particles collide, but only the fraction of collisions possessing kinetic energy greater than or equal to the activation energy ($E \geq E_a$) and correct spatial orientation can successfully rearrange bonds to form products.	[1]
18	A	Option A is correct. Collision frequency determines how often particles collide, but only the fraction of collisions possessing kinetic energy greater than or equal to the activation energy ($E \geq E_a$) and correct spatial orientation can successfully rearrange bonds to form products.	[1]

24

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
39	D	Option D is correct. Collision frequency determines how often particles collide, but only the fraction of collisions possessing kinetic energy greater than or equal to the activation energy ($E \geq E_a$) and correct spatial orientation can successfully rearrange bonds to form products.	[1]
40	B	Option B is correct. Collision frequency determines how often particles collide, but only the fraction of collisions possessing kinetic energy greater than or equal to the activation energy ($E \geq E_a$) and correct spatial orientation can successfully rearrange bonds to form products.	[1]
41	B	Option B is correct. A Maxwell-Boltzmann distribution curve: (1) begins at the origin because no molecules have zero energy, (2) is asymmetrical (skewed to the right), (3) its peak corresponds to the most probable energy (E_{mp}), and (4) at high energies, the curve tails off asymptotically towards the x-axis but never touches it because there is no theoretical maximum energy for a particle.	[1]
42	D	Option D is correct. At higher temperature T_2 , average molecular kinetic energy increases, shifting the peak (most probable energy) to the right. Because total molecules are unchanged, the total area under the curve is strictly conserved (area = total particles). To accommodate the broader distribution of molecules with higher energies, the curve flattens and the peak height decreases.	[1]
43	A	Option A is correct. A 10 K temperature rise increases average molecular velocity by only ~1.7%, increasing collision frequency by less than 2% (which would produce a negligible rate increase). The primary reason is that in the exponential tail of the Maxwell-Boltzmann distribution, even a slight shift of the curve to the right dramatically multiplies the number of molecules possessing energy $\geq E_a$, roughly doubling the number of successful collisions per second.	[1]
44	C	Option C is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
45	A	Option A is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
46	D	Option D is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
47	B	Option B is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
48	C	Option C is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
49	B	Option B is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
50	A	Option A is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
51	D	Option D is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
52	C	Option C is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
53	A	Option A is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
54	C	Option C is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
55	B	Option B is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
56	D	Option D is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
57	C	Option C is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
58	A	Option A is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
59	D	Option D is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
60	B	Option B is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
61	B	Option B is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
62	D	Option D is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
63	A	Option A is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
64	C	Option C is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
65	A	Option A is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
66	D	Option D is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
67	B	Option B is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
68	C	Option C is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
69	B	Option B is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
70	A	Option A is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
71	D	Option D is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
72	C	Option C is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
73	A	Option A is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
74	C	Option C is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
75	B	Option B is correct. The y-axis of a Maxwell-Boltzmann distribution plot represents the number of molecules per unit energy interval. Therefore, the integral (area under the curve) from energy $E = 0$ to infinity equals the total number of gas molecules in the system.	[1]
76	D	Option D is correct. A catalyst does not increase temperature or kinetic energy. Instead, it provides a different molecular reaction mechanism (e.g. via an intermediate or surface adsorption) that has a lower activation energy barrier ($E_a(\text{cat}) < E_a$). As a result, a much larger fraction of molecules possess energy $> E_a(\text{cat})$, accelerating the reaction rate.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
77	C	Option C is correct. Adding a catalyst does not alter the temperature, so the shape and position of the Maxwell-Boltzmann distribution curve remain completely unchanged. The catalyst lowers the required activation energy threshold, shifting the E_a line to the left. The area under the curve to the right of this new threshold (representing molecules with sufficient energy to react) is significantly enlarged.	[1]
78	A	Option A is correct. A homogeneous catalyst exists in the same physical phase as the reactants (for example, aqueous Fe^{2+} ions catalysing a reaction between aqueous peroxodisulfate and iodide ions, or gaseous NO_2 catalysing the gas-phase oxidation of SO_2).	[1]
79	D	Option D is correct. The uncatalysed reaction requires two negatively charged anions ($\text{S}_2\text{O}_8^{2-}$ and I^-) to collide, which has a very high activation energy due to strong electrostatic repulsion between like charges. $\text{Fe}^{2+}/\text{Fe}^{3+}$ ions have variable oxidation states and act as a redox bridge: Step 1: $2\text{Fe}^{2+} + \text{S}_2\text{O}_8^{2-} \rightarrow 2\text{Fe}^{3+} + 2\text{SO}_4^{2-}$ (attraction between positive Fe^{2+} and negative $\text{S}_2\text{O}_8^{2-}$). Step 2: $2\text{Fe}^{3+} + 2\text{I}^- \rightarrow 2\text{Fe}^{2+} + \text{I}_2$ (attraction between positive Fe^{3+} and negative I^-). Both steps involve attractive oppositely charged ions with much lower activation energies.	[1]
80	B	Option B is correct. Heterogeneous catalysis proceeds through four sequential steps: (1) Adsorption: gas molecules bond chemically (chemisorption) to active sites on the solid catalyst surface; (2) Bond weakening: electron donation/acceptance weakens internal bonds in reactants, lowering E_a ; (3) Reaction: adsorbed species react to form product molecules; (4) Desorption: products detach from active sites and diffuse away into the gas phase, regenerating the active sites.	[1]
81	B	Option B is correct. Catalyst poisoning occurs when an impurity (such as lead compounds or sulfur dioxide) forms strong, irreversible chemical bonds with the active sites on the catalyst surface. Because these poison molecules do not desorb, they permanently block reactant molecules from adsorbing, rendering the catalyst inactive.	[1]
82	D	Option D is correct. Catalyst poisoning occurs when an impurity (such as lead compounds or sulfur dioxide) forms strong, irreversible chemical bonds with the active sites on the catalyst surface. Because these poison molecules do not desorb, they permanently block reactant molecules from adsorbing, rendering the catalyst inactive.	[1]
83	A	Option A is correct. Catalyst poisoning occurs when an impurity (such as lead compounds or sulfur dioxide) forms strong, irreversible chemical bonds with the active sites on the catalyst surface. Because these poison molecules do not desorb, they permanently block reactant molecules from adsorbing, rendering the catalyst inactive.	[1]
84	C	Option C is correct. Catalyst poisoning occurs when an impurity (such as lead compounds or sulfur dioxide) forms strong, irreversible chemical bonds with the active sites on the catalyst surface. Because these poison molecules do not desorb, they permanently block reactant molecules from adsorbing, rendering the catalyst inactive.	[1]
85	A	Option A is correct. Catalyst poisoning occurs when an impurity (such as lead compounds or sulfur dioxide) forms strong, irreversible chemical bonds with the active sites on the catalyst surface. Because these poison molecules do not desorb, they permanently block reactant molecules from adsorbing, rendering the catalyst inactive.	[1]
86	D	Option D is correct. Catalyst poisoning occurs when an impurity (such as lead compounds or sulfur dioxide) forms strong, irreversible chemical bonds with the active sites on the catalyst surface. Because these poison molecules do not desorb, they permanently block reactant molecules from adsorbing, rendering the catalyst inactive.	[1]
87	B	Option B is correct. Catalyst poisoning occurs when an impurity (such as lead compounds or sulfur dioxide) forms strong, irreversible chemical bonds with the active sites on the catalyst surface. Because these poison molecules do not desorb, they permanently block reactant molecules from adsorbing, rendering the catalyst inactive.	[1]
88	C	Option C is correct. Catalyst poisoning occurs when an impurity (such as lead compounds or sulfur dioxide) forms strong, irreversible chemical bonds with the active sites on the catalyst surface. Because these poison molecules do not desorb, they permanently block reactant molecules from adsorbing, rendering the catalyst inactive.	[1]
89	B	Option B is correct. Catalyst poisoning occurs when an impurity (such as lead compounds or sulfur dioxide) forms strong, irreversible chemical bonds with the active sites on the catalyst surface. Because these poison molecules do not desorb, they permanently block reactant molecules from adsorbing, rendering the catalyst inactive.	[1]
90	A	Option A is correct. Catalyst poisoning occurs when an impurity (such as lead compounds or sulfur dioxide) forms strong, irreversible chemical bonds with the active sites on the catalyst surface. Because these poison molecules do not desorb, they permanently block reactant molecules from adsorbing, rendering the catalyst inactive.	[1]
91	D	Option D is correct. Catalyst poisoning occurs when an impurity (such as lead compounds or sulfur dioxide) forms strong, irreversible chemical bonds with the active sites on the catalyst surface. Because these poison molecules do not desorb, they permanently block reactant molecules from adsorbing, rendering the catalyst inactive.	[1]
92	C	Option C is correct. Catalyst poisoning occurs when an impurity (such as lead compounds or sulfur dioxide) forms strong, irreversible chemical bonds with the active sites on the catalyst surface. Because these poison molecules do not desorb, they permanently block reactant molecules from adsorbing, rendering the catalyst inactive.	[1]
93	A	Option A is correct. Catalyst poisoning occurs when an impurity (such as lead compounds or sulfur dioxide) forms strong, irreversible chemical bonds with the active sites on the catalyst surface. Because these poison molecules do not desorb, they permanently block reactant molecules from adsorbing, rendering the catalyst inactive.	[1]
94	C	Option C is correct. Catalyst poisoning occurs when an impurity (such as lead compounds or sulfur dioxide) forms strong, irreversible chemical bonds with the active sites on the catalyst surface. Because these poison molecules do not desorb, they permanently block reactant molecules from adsorbing, rendering the catalyst inactive.	[1]
95	B	Option B is correct. Catalyst poisoning occurs when an impurity (such as lead compounds or sulfur dioxide) forms strong, irreversible chemical bonds with the active sites on the catalyst surface. Because these poison molecules do not desorb, they permanently block reactant molecules from adsorbing, rendering the catalyst inactive.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
96	D	Option D is correct. Catalyst poisoning occurs when an impurity (such as lead compounds or sulfur dioxide) forms strong, irreversible chemical bonds with the active sites on the catalyst surface. Because these poison molecules do not desorb, they permanently block reactant molecules from adsorbing, rendering the catalyst inactive.	[1]
97	C	Option C is correct. Catalyst poisoning occurs when an impurity (such as lead compounds or sulfur dioxide) forms strong, irreversible chemical bonds with the active sites on the catalyst surface. Because these poison molecules do not desorb, they permanently block reactant molecules from adsorbing, rendering the catalyst inactive.	[1]
98	A	Option A is correct. Catalyst poisoning occurs when an impurity (such as lead compounds or sulfur dioxide) forms strong, irreversible chemical bonds with the active sites on the catalyst surface. Because these poison molecules do not desorb, they permanently block reactant molecules from adsorbing, rendering the catalyst inactive.	[1]
99	D	Option D is correct. Catalyst poisoning occurs when an impurity (such as lead compounds or sulfur dioxide) forms strong, irreversible chemical bonds with the active sites on the catalyst surface. Because these poison molecules do not desorb, they permanently block reactant molecules from adsorbing, rendering the catalyst inactive.	[1]
100	B	Option B is correct. Catalyst poisoning occurs when an impurity (such as lead compounds or sulfur dioxide) forms strong, irreversible chemical bonds with the active sites on the catalyst surface. Because these poison molecules do not desorb, they permanently block reactant molecules from adsorbing, rendering the catalyst inactive.	[1]
101	C	Option C is correct. Increasing temperature increases the average kinetic energy of gas molecules, shifting the peak of the distribution to the right. To keep total area constant (fixed number of molecules), the peak height must decrease and the curve flattens. Most crucially, the area under the tail of the curve to the right of the activation energy threshold ($E > E_a$) expands dramatically, explaining the marked increase in reaction rate.	[1]
102	A	Option A is correct. A catalyst provides an alternative pathway with a lower activation energy. It does not alter the temperature or the distribution of molecular kinetic energies, so the curve itself is unchanged. The activation energy threshold line is relocated to the left (lower energy), resulting in a larger fraction of molecules possessing sufficient energy to react.	[1]
103	D	Option D is correct. A 10 °C temperature increase (e.g. from 300 K to 310 K) increases absolute temperature by only ~3%, so collision frequency increases by less than 2%. The exponential dependence of rate on temperature arises because even a slight rightward shift of the Maxwell-Boltzmann distribution roughly doubles the number of molecules possessing energy in the high-energy tail beyond the activation energy barrier E_a .	[1]
104	B	Option B is correct. In heterogeneous catalysis: (1) Adsorption: gaseous reactant molecules bind to active sites on the solid catalyst surface; (2) Bond weakening: interaction with the surface weakens internal bonds of the reactants, lowering activation energy; (3) Reaction: adsorbed species react to form products; (4) Desorption: product molecules release from the active sites, allowing fresh reactants to bind.	[1]
105	A	Option A is correct. The uncatalysed reaction involves collision between two negative ions ($\text{S}_2\text{O}_8^{2-}$ and I^-), which is hindered by electrostatic repulsion. The catalyst Fe^{3+} provides an alternative two-step route involving collisions between oppositely charged ions: Step 1: $2\text{Fe}^{3+}(\text{aq}) + 2\text{I}^-(\text{aq}) \rightarrow 2\text{Fe}^{2+}(\text{aq}) + \text{I}_2(\text{aq})$. Step 2: $2\text{Fe}^{2+}(\text{aq}) + \text{S}_2\text{O}_8^{2-}(\text{aq}) \rightarrow 2\text{Fe}^{3+}(\text{aq}) + 2\text{SO}_4^{2-}(\text{aq})$. Iron(III) is regenerated unchanged at the end.	[1]
106	C	Option C is correct. The rate of reaction at any moment is given by the gradient of the concentration–time curve ($d[\text{reactant}]/dt$). The initial rate is the instantaneous rate at the beginning of the reaction ($t = 0$), obtained by drawing a tangent line to the curve at $t = 0$ and determining the magnitude of its slope (gradient = $\Delta[\text{concentration}] / \Delta[\text{time}]$).	[1]
107	B	Option B is correct. Aqueous bromine is deeply colored (orange–brown), whereas the products (bromide ions, hydrogen ions, carbon dioxide, water) and methanoic acid are all completely colorless. As the reaction proceeds, the absorbance of light decreases continuously in direct proportion to the concentration of remaining Br_2 , making colorimetry the most sensitive, continuous, and accurate method.	[1]
108	D	Option D is correct. In catalytic converters, toxic carbon monoxide (CO , a poisonous gas) and harmful nitrogen monoxide (NO , a contributor to acid rain and photochemical smog) are converted into harmless carbon dioxide (CO_2) and unreactive atmospheric nitrogen gas (N_2) over a platinum/rhodium heterogeneous catalyst: $2\text{CO} + 2\text{NO} \rightarrow 2\text{CO}_2 + \text{N}_2$.	[1]
109	A	Option A is correct. Leaded petrol contains tetraethyllead as an antiknock additive. During combustion, lead oxides are formed that bind strongly and irreversibly to the active transition metal sites (Pt , Pd , Rh) on the catalyst surface. Because lead does not desorb, it permanently blocks the active sites from interacting with exhaust gases (catalyst poisoning).	[1]
110	C	Option C is correct. A catalyst provides an alternative reaction pathway that lowers the activation energy barrier (E_a) for both the forward and reverse reactions by the exact same amount. Because the initial chemical potential energy of the reactants and the final chemical potential energy of the products are intrinsic thermodynamic properties of the substances, the overall enthalpy change of reaction ($\Delta H = H_{\text{products}} - H_{\text{reactants}}$) is completely unaltered.	[1]